

Lab Assignment no. 2

PRACTICAL NO. 3

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SUBJECT: EDS LAB

3.1.1.NUMPY ARRAY OPERATIONS

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.

- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, r and c , representing the number of rows and the number of columns.
- The next r lines contain c space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Code:

```
import numpy as np

rows,columns=map(int,input().split()) elements=[]

for i in range (rows):

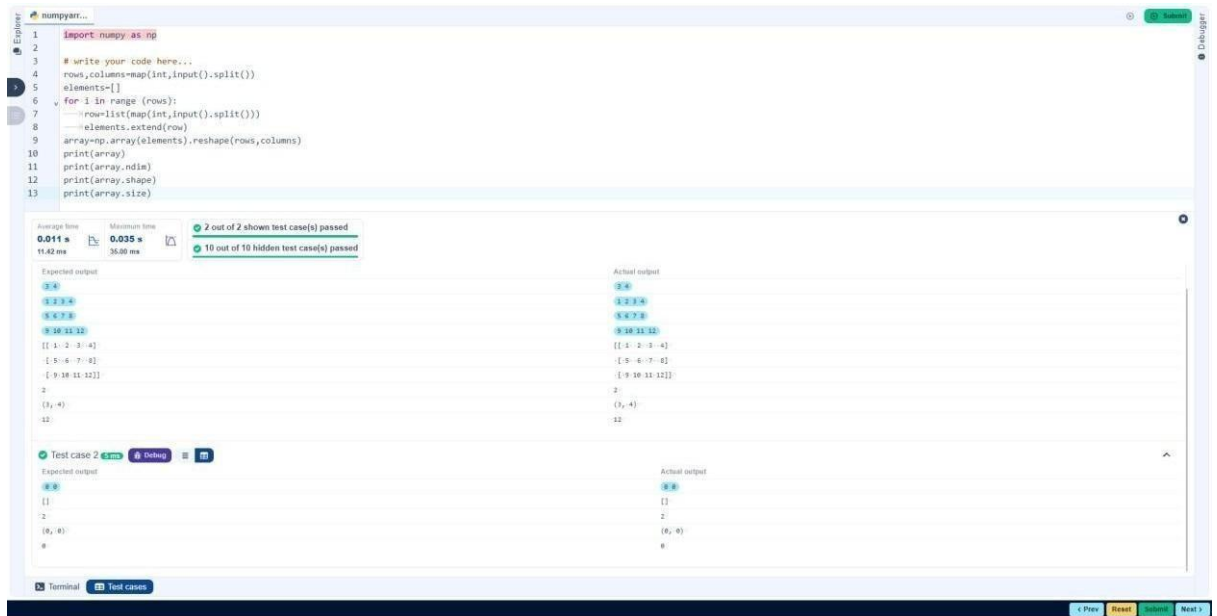
row=list(map(int,input().split()))

elements.extend(row)

array=np.array(elements).reshape(rows,columns)
```

```
print(array) print(array.ndim) print(array.shape)

print(array.size)
```



3.2.1. NUMPY: MATRIX OPERATIONS

The given code takes two 3 x 3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. Addition (`matrix_a + matrix_b`)
2. Subtraction (`matrix_a - matrix_b`)
3. Element-wise Multiplication (`matrix_a * matrix_b`)
4. Matrix Multiplication (`matrix_a . matrix_b`)
5. Transpose of Matrix A

Input Format:

- The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.

Similarly, the user will input 3 rows for matrix_b, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. **The result of Addition.**
2. **The result of Subtraction.**
3. **The result of Element-wise Multiplication.**
4. **The result of Matrix Multiplication.**
5. **The Transpose of Matrix A.**

Code:

```
import numpy as np
```

```
# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])
```

```
# Addition
print("Addition (A + B):")
```

```
add=np.array(matrix_a+matrix_b)
print(add)
```

```
# Subtraction
print("Subtraction (A - B):")
```

```
sub=np.array(matrix_a-matrix_b)
print(sub)
```

```
# Multiplication (element-wise)
print("Elementwise Multiplication (A * B):")
```

```

multi=np.array(matrix_a*matrix_b) print(multi) # Matrix
multiplication (dot product) print("A dot B:")

trans=(matrix_a.dot(matrix_b))    print(trans)    #
Transpose    print("Transpose    of    A:")

tran=matrix_a.transpose() print(tran)

```



3.2.2.NUMPY: HORIZONTAL VS VERTICAL STACKING OF ARRAYS

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking:** Stack the two matrices horizontally (side by side).

-

Vertical Stacking: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Code:

```
import numpy as np
```

```
# Input matrices print("Enter Array1:") arr1 = np.array([list(map(int,
input().split())) for i in range(3)])
```

```
print("Enter Array2:") arr2 = np.array([list(map(int, input().split()))
for i in range(3)])
```

```
# Perform horizontal stacking (hstack) print("Horizontal
Stack:") print(np.hstack((arr1,arr2))) print("Vertical
Stack:")
```

```

stacking.py
1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Array2:")
8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10 # Perform horizontal stacking (hstack)
11 print("Horizontal Stack:")
12 print(np.hstack((arr1, arr2)))

```

Average time: 0.022 s, Maximum time: 0.033 s, 2 out of 2 shown test case(s) passed, 2 out of 2 hidden test case(s) passed

Test case 1 33 ms

Expected output	Actual output
Enter Array1:	Enter Array1:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Array2:	Enter Array2:
4 5 6	4 5 6
7 8 9	7 8 9
10 11 12	10 11 12
Horizontal Stack:	Horizontal Stack:
[[1 2 3 4 5 6]]	[[1 2 3 4 5 6]]
[-4 -5 -6 -7 -8 -9]	[-4 -5 -6 -7 -8 -9]
[-7 -8 -9 10 11 12]]	[-7 -8 -9 10 11 12]]
Vertical Stack:	Vertical Stack:
[[1 2 3]]	[[1 2 3]]
[-4 -5 -6]	[-4 -5 -6]
[-7 -8 -9]	[-7 -8 -9]
[-4 -5 -6]	[-4 -5 -6]
[-7 -8 -9]	[-7 -8 -9]
[-10 -11 -12]]	[-10 -11 -12]]

```

stacking.py
1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Array2:")
8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10 # Perform horizontal stacking (hstack)
11 print("Horizontal Stack:")
12 print(np.hstack((arr1, arr2)))

```

Average time: 0.022 s, Maximum time: 0.033 s, 2 out of 2 shown test case(s) passed, 2 out of 2 hidden test case(s) passed

Test case 2 21 ms

Expected output	Actual output
Enter Array1:	Enter Array1:
-1 -2 -3	-1 -2 -3
-4 -5 -6	-4 -5 -6
-7 -8 -9	-7 -8 -9
Enter Array2:	Enter Array2:
10 11 12	10 11 12
13 14 15	13 14 15
16 17 18	16 17 18
Horizontal Stack:	Horizontal Stack:
[[-1 -2 -3 10 11 12]]	[[-1 -2 -3 10 11 12]]
[-4 -5 -6 13 14 15]	[-4 -5 -6 13 14 15]
[-7 -8 -9 16 17 18]]	[-7 -8 -9 16 17 18]]
Vertical Stack:	Vertical Stack:
[[-1 -2 -3]]	[[-1 -2 -3]]
[-4 -5 -6]	[-4 -5 -6]
[-7 -8 -9]	[-7 -8 -9]
[10 11 12]	[10 11 12]
[13 14 15]	[13 14 15]
[16 17 18]]	[16 17 18]]

3.2.3. NUMPY: CUSTOM SEQUENCE GENERATION

Write a Python program that takes the following inputs from the user:

- **Start value:** The starting point of the sequence.
- **Stop value:** The sequence should end before this value.
- **Step value:** The increment between each number in the sequence.

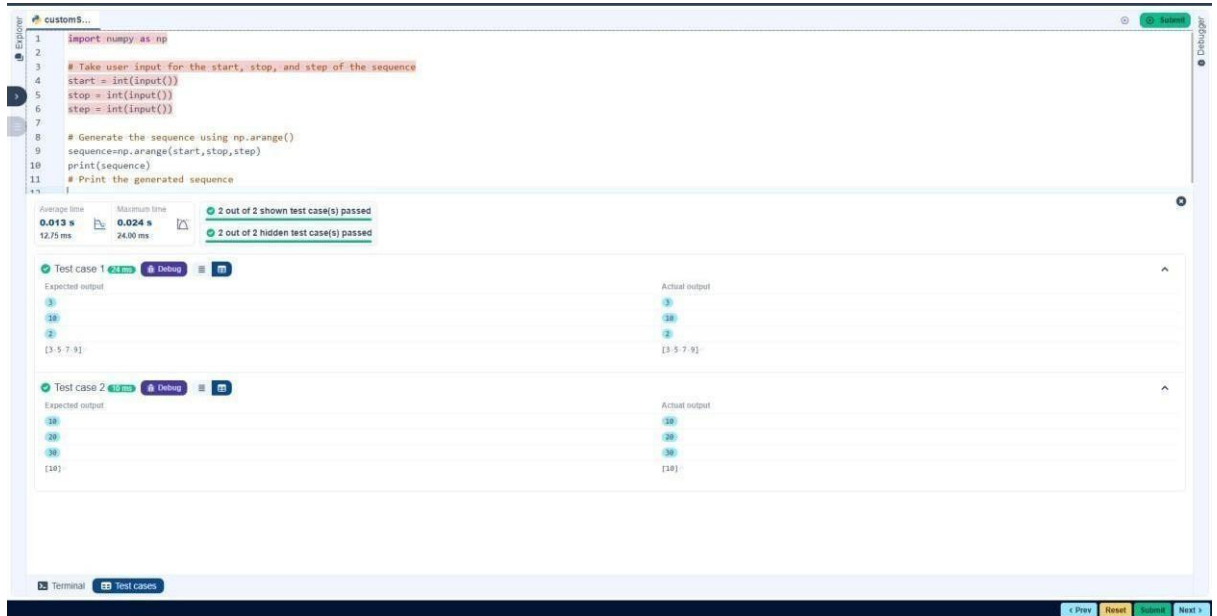
The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

The program should print the generated sequence based on the input values.



3.2.4. NUMPY: ARITHMETIC AND STATISTICAL OPERATIONS, MATHEMATICAL OPERATIONS, BITWISE OPERATORS

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: $A_i \text{ OR } B_i$).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Code:

```
import numpy as np
```

```
def array_operations(A, B):
```

```
    # Convert A and B to NumPy
```

```
    arrays x=np.array(A) y=np.array(B) #
```

```
    Arithmetic Operations          sum_result
```

```
= x+y  diff_result = x-y          prod_result
```

```
= x*y
```

```
    # Statistical Operations
```

```
    mean_A = np.mean(A)  median_A
```

```
= np.median(A)  std_dev_A =
```

```
    np.std(A)
```

```
    # Bitwise Operations  and_result =
```

```
    np.bitwise_and(x,y)  or_result = np.bitwise_or(x,y)
```

```
    xor_result
```

```
•  
= np.bitwise_xor(x,y)  
  
# Output results with one space between each element print("Elementwise  
Sum:", ' '.join(map(str, sum_result))) print("Element-wise Difference:", ' '.join(map(str,  
diff_result))) print("Element-wise Product:", ' '.join(map(str, prod_result)))
```

```

print(f"Mean of A: {mean_A}") print(f"Median of
A: {median_A}") print(f"Standard
Deviation of A: {std_dev_A}")

print("Bitwise AND:", ' '.join(map(str, and_result))) print("Bitwise
OR:", ' '.join(map(str, or_result))) print("Bitwise XOR:", '
'.join(map(str, xor_result)))

A = list(map(int, input().split())) # Elements of array A
B = list(map(int, input().split())) # Elements of array B

array_operations(A, B)

```

The screenshot shows the CodeTANTRA IDE interface. The editor displays a Python script with the following code:

```

11 prod_result = x*y
12
13 # Statistical Operations
14 mean_A = np.mean(A)
15 median_A = np.median(A)
16 std_dev_A = np.std(A)
17
18 # Bitwise Operations
19 and_result = np.bitwise_and(x,y)
20 or_result = np.bitwise_or(x,y)
21 xor_result = np.bitwise_xor(x,y)
22
23 # Output results with one space between each element
24 print("Element-wise Sum:", ' '.join(map(str, sum_result)))
25 print("Element-wise Difference:", ' '.join(map(str, diff_result)))
26 print("Element-wise Product:", ' '.join(map(str, prod_result)))
27
28 print(f"Mean of A: {mean_A}")

```

Below the editor, the test results are shown. The 'Test case 1' section indicates that 1 out of 1 shown test case(s) passed. The 'Expected output' and 'Actual output' sections both show the following results:

```

Element-wise Sum: 6 9 10 12
Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32
Mean of A: 2.5
Median of A: 2.5
Standard Deviation of A: 1.118033988749895
Bitwise AND: 1 2 3 0
Bitwise OR: 5 6 7 12
Bitwise XOR: 4 4 4 12

```

3.2.5.NUMPY: COPYING AND VIEWING ARRAYS

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

-

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>

View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>

Copy array: <copy_array>

Code: `import numpy`

`as np`

```
inputlist = list(map(int,input().split(" ")))
```

```
# Original array original_array = np.array(inputlist)
```

```
# Create a view view_array
```

```
=original_array.view()
```

```
# Create a copy copy_array
```

```
=original_array.copy()
```

```
# Modify the view view_array[0] = 99 print("Original array after  
modifying view:", original_array) print("View array:", view_array)
```

```
# Modify the copy copy_array[1] = 88 print("Original array after
modifying copy:", original_array) print("Copy array:",
copy_array)
```

The screenshot shows the CodeTANTRA IDE interface. The editor displays a Python script for array operations using NumPy. The script takes input as a space-separated string, converts it to a NumPy array, creates a view and a copy, and then modifies the view. The test results panel shows two test cases passed, with expected and actual outputs for each.

```
1 import numpy as np
2
3 inputlist = list(map(int, input().split(" ")))
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array = original_array.view()
10
11 # Create a copy
12 copy_array = original_array.copy()
13
14 # Modify the view
15 view_array[0] = 99
```

Test Results:

- 2 out of 2 shown test case(s) passed
- 2 out of 2 hidden test case(s) passed

Test case 1:

Expected output	Actual output
99 20 30 40 50 60 70 80	99 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]	Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]	View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]	Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 80 30 40 50 60 70 80]	Copy array: [10 80 30 40 50 60 70 80]

Test case 2:

Expected output	Actual output
99 2 3 4 5	99 2 3 4 5
Original array after modifying view: [99 -2 -3 -4 -5]	Original array after modifying view: [99 -2 -3 -4 -5]
View array: [99 -2 -3 -4 -5]	View array: [99 -2 -3 -4 -5]
Original array after modifying copy: [99 -2 -3 -4 -5]	Original array after modifying copy: [99 -2 -3 -4 -5]
Copy array: [99 88 -3 -4 -5]	Copy array: [99 88 -3 -4 -5]

3.2.6. NUMPY: SEARCHING, SORTING, COUNTING, BROADCASTING

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- **search_value:** The value to search for in the array.
- **count_value:** The value to count its occurrences in the array.
- **broadcast_value:** The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. **Searching:** Find the indices where search_value appears in array1 and print these indices.
2. **Counting:** Count how many times count_value appears in array1 and print the count.

3. **Broadcasting:** Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
4. **Sorting:** Sort `array1` in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing `array1`.
2. An integer `search_value` represents the value to search for in the array.
3. An integer `count_value` represents the value to count in the array.
4. An integer `broadcast_value` represents the value to add to each element of the array.

Output Format:

1. The indices where `search_value` occurs in `array1`.
2. The count of occurrences of `count_value` in `array1`.
3. The array after adding the `broadcast_value` to each element.
4. The sorted array.

Code:

```
import numpy as np
```

```
# Input array from the user array1 = np.array(list(map(int,
input().split())))
```

```
# Searching search_value = int(input("Value to
search: ")) count_value = int(input("Value to
count: ")) broadcast_value = int(input("Value to
add: "))
```

```

# Find indices where value matches in array1
a=np.where(array1==search_value)[0]

print(a)

# Count occurrences in arraycount_nonzero(array1==count_value)

b=np.count_nonzero(array1==count_value)

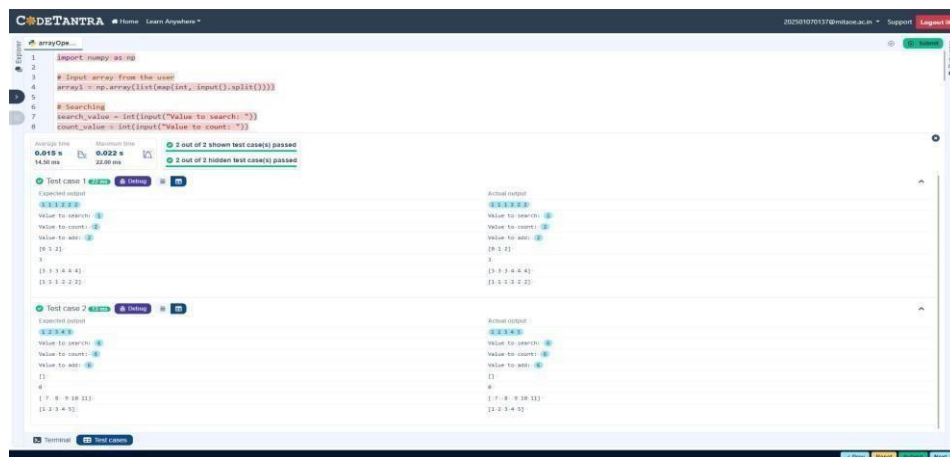
print(b)
# Broadcasting addition c=array1

+ broadcast_value print(c)

# Sort the first array

d=np.sort(array1) print(d)

```



3.2.7.STUDENT DATA ANALYSIS AND OPERATIONS

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.

- Find maximum marks in Subject 3: Identify the highest marks in Subject 3.
- Print all subject marks: Display the marks of all students for each subject.
- Find total marks of students: Compute the total marks for each student across all subjects.
- Find the average marks of each student: Compute the average marks for each student.
- Find average marks of each subject: Compute the average marks for all students in each subject.
- Find average marks of Subject 1 and Subject 2: Compute the average marks for Subject 1 and Subject 2.
- Find average marks of Subject 1 and Subject 3: Compute the average marks for Subject 1 and Subject 3.
- Find the roll number of the student with maximum marks in Subject 3: Identify the student with the highest marks in Subject 3 and print their roll number.
- Find the roll number of the student with minimum marks in Subject 2: Identify the student with the lowest marks in Subject 2 and print their roll number.
- Find the roll number of students who scored 24 marks in Subject 2: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- Find the count of students who got less than 40 marks in Subject 1: Count the number of students who scored less than 40 marks in Subject 1.
- Find the count of students who got more than 90 marks in Subject 2: Count the number of students who scored more than 90 marks in Subject 2.
- Find the count of students who scored ≥ 90 in each subject: Count the number of students who scored 90 or more marks in each subject.
- Find the count of subjects in which each student scored ≥ 90 : Determine how many subjects each student scored 90 or more marks in.
- Print Subject 1 marks in ascending order: Sort and print the marks of students in Subject 1 in ascending order.
- Print students who scored between 50 and 90 in Subject 1: Display students who scored marks between 50 and 90 in Subject 1.
- Find index positions of students who scored 79 in Subject 1: Identify the index positions of students who scored exactly 79 marks in Subject 1.

Code:

```
import numpy as np
```

```
a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
```

```
# 1. Print all student details print("All student  
Details:\n", a )
```

```
# 2. print total students
```

```
print("Total Students:",a.shape[0] )
```

```
# 3. Print all student Roll numbers print("All  
Student Roll Nos", a[:,0] )
```

```
# 4. Print subject 1 marks print("Subject  
1 Marks", a[:,1])
```

```
# 5. print minimum marks of Subject 2 print("Min marks  
in Subject 2", np.min(a[:,2] ))
```

```
# 6. print maximum marks of Subject 3 print("Max marks in  
Subject 3", np.max(a[:,3]) )
```

```
# 7. Print All subject marks print("All subject  
marks:", a[:,1:])
```

```
# 8. print Total marks of students print("Total  
Marks", np.sum(a[:,1:],axis=1) )
```

```
# 9. print average marks of each student avg=np.mean(a[:,1:],axis=1)  
print(np.round(avg,1))
```

```
# 10. print average marks of each subject print("Average  
Marks of each subject", np.mean(a[:,1:],axis=0) )
```

```
# 11. print average marks of S1 and S2 print("Average Marks of  
S1 and S2", np.mean(a[:,1:3],axis=0) )
```

```
# 12. print average marks of S1 and S3 print("Average Marks of  
S1 and S3", np.mean(a[:,[1,3]],axis=0) )
```

```
# 13. print Roll number who got maximum marks in Subject 3 i=np.argmax(a[:,3]) print("Roll no  
who got maximum marks in Subject 3", a[i,0] )
```

```
# 14. print Roll number who got minimum marks in Subject 2  
j=np.argmin(a[:,2]) print("Roll no who got minimum marks in Subject  
2", a[j,0] )
```

```
# 15. print Roll number who got 24 marks in Subject 2  
  
print(f"Roll no who got 24 marks in Subject 2 [{a[a[:,2]==24,0}]" )
```

```
# 16. print count of students who got marks in Subject 1 < 40 print("Count of  
students who got marks in Subject 1 < 40", np.sum(a[:,1]<40) )
```

```

# 17. print count of students who got marks in Subject 2 > 90 print("Count of
students who got marks in Subject 2 > 90:", np.sum(a[:,2]>90))

Count=[len(np.argwhere(a[:,1]>=90)),len(np.argwhere(a[:,2]>=90)),len(np.argwhere(a[:,3]>= 90))]
count=np.array(Count)

# 18. print count of students in each subject who got marks >= 90 print("Count of
students in each subject who got marks >= 90:", count)

# 19. print count of subjects in which each student got marks >= 90 print("Roll no:", a[:,0]
) print("Count of subjects in which student got marks >= 90:", np.sum(a[:,1:]>=90,axis=1)
)

```

```
(a[:,1]<=90])) k=np.where(a[:,1]==79)
```

```
print(k)
```

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Operation...

```
45 print("Roll no who got maximum marks in Subject 3", a[i,0].....)
46
47 # 14. print Roll number who got minimum marks in Subject 2
48 j=np.argmin(a[:,2])
49 print("Roll no who got minimum marks in Subject 2", a[j,0]...)
```

Average time 0.007 s Maximum time 0.007 s
7.00 ms 7.00 ms

1 out of 1 shown test case(s) passed

Test case 1 7 ms Debug

Expected output	Actual output
All student Details:	All student Details:
[[301, 67, 77, 88]]	[[301, 67, 77, 88]]
[[302, 78, 88, 77]]	[[302, 78, 88, 77]]
[[303, 45, 56, 89]]	[[303, 45, 56, 89]]
[[304, 88, 98, 45]]	[[304, 88, 98, 45]]
[[305, 78, 88, 99]]	[[305, 78, 88, 99]]
[[306, 68, 78, 36]]	[[306, 68, 78, 36]]
[[307, 79, 12, 45]]	[[307, 79, 12, 45]]
[[308, 46, 45, 77]]	[[308, 46, 45, 77]]
[[309, 89, 32, 88]]	[[309, 89, 32, 88]]
[[310, 79, 24, 33]]	[[310, 79, 24, 33]]
Total Students: 10	Total Students: 10
All Student Roll Nos [301, 302, 303, 304, 305, 306, 307, 308, 309, 310]	All Student Roll Nos [301, 302, 303, 304, 305, 306, 307, 308, 309, 310]
Subject 1 Marks [67, 78, 45, 88, 78, 68, 79, 46, 89, 79]	Subject 1 Marks [67, 78, 45, 88, 78, 68, 79, 46, 89, 79]
Min marks in Subject 2 12.0	Min marks in Subject 2 12.0
Max marks in Subject 3 99.0	Max marks in Subject 3 99.0
All subject marks: [[67, 77, 88]]	All subject marks: [[67, 77, 88]]
[[78, 88, 77]]	[[78, 88, 77]]
[[45, 56, 89]]	[[45, 56, 89]]
[[88, 98, 45]]	[[88, 98, 45]]
[[78, 88, 99]]	[[78, 88, 99]]
[[68, 78, 36]]	[[68, 78, 36]]
[[79, 12, 45]]	[[79, 12, 45]]
[[46, 45, 77]]	[[46, 45, 77]]
[[89, 32, 88]]	[[89, 32, 88]]
[[79, 24, 33]]	[[79, 24, 33]]
Total Marks [232, 243, 190, 231, 265, 182, 136, 168, 209, 136]	Total Marks [232, 243, 190, 231, 265, 182, 136, 168, 209, 136]
[77.3 81. 63.3 77. 88.3 68.7 45.3 56. 69.7 45.3]	[77.3 81. 63.3 77. 88.3 68.7 45.3 56. 69.7 45.3]
Average Marks of each subject [71.7 59.8 67.7]	Average Marks of each subject [71.7 59.8 67.7]
Average Marks of S1 and S2 [71.7 59.8]	Average Marks of S1 and S2 [71.7 59.8]
Average Marks of S1 and S3 [71.7 67.7]	Average Marks of S1 and S3 [71.7 67.7]

Terminal Test cases

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Operation...

```
43 # 13. print Roll number who got maximum marks in Subject 3
44 i=np.argmax(a[:,3])
45 print("Roll no who got maximum marks in Subject 3", a[i,0].....)
```

Average time 0.007 s Maximum time 0.007 s
7.00 ms 7.00 ms

1 out of 1 shown test case(s) passed

Average Marks of S1 and S3 [71.7 67.7]	Average Marks of S1 and S3 [71.7 67.7]
Roll no who got maximum marks in Subject 3 305.0	Roll no who got maximum marks in Subject 3 305.0
Roll no who got minimum marks in Subject 2 307.0	Roll no who got minimum marks in Subject 2 307.0
Roll no who got 24 marks in Subject 2 [[310.]]	Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40 0	Count of students who got marks in Subject 1 < 40 0
Count of students who got marks in Subject 2 > 90: 1	Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0 1 1]	Count of students in each subject who got marks >= 90: [0 1 1]
Roll no: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310]	Roll no: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]	Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]
[45, 46, 67, 68, 78, 78, 79, 79, 88, 89]	[45, 46, 67, 68, 78, 78, 79, 79, 88, 89]
[[301, 67, 77, 88]]	[[301, 67, 77, 88]]
[[302, 78, 88, 77]]	[[302, 78, 88, 77]]
[[304, 88, 98, 45]]	[[304, 88, 98, 45]]
[[305, 78, 88, 99]]	[[305, 78, 88, 99]]
[[306, 68, 78, 36]]	[[306, 68, 78, 36]]
[[307, 79, 12, 45]]	[[307, 79, 12, 45]]
[[309, 89, 32, 88]]	[[309, 89, 32, 88]]
[[310, 79, 24, 33]]	[[310, 79, 24, 33]]
[[301, 67, 77, 88]]	[[301, 67, 77, 88]]
[[302, 78, 88, 77]]	[[302, 78, 88, 77]]
[[303, 45, 56, 89]]	[[303, 45, 56, 89]]
[[304, 88, 98, 45]]	[[304, 88, 98, 45]]
[[305, 78, 88, 99]]	[[305, 78, 88, 99]]
[[306, 68, 78, 36]]	[[306, 68, 78, 36]]
[[307, 79, 12, 45]]	[[307, 79, 12, 45]]
[[308, 46, 45, 77]]	[[308, 46, 45, 77]]
[[309, 89, 32, 88]]	[[309, 89, 32, 88]]
[[310, 79, 24, 33]]	[[310, 79, 24, 33]]
(array([6, 9], dtype=int32),)	(array([6, 9], dtype=int32),)

Terminal Test cases

```
# 20. Print S1 marks in ascending order print(np.sort(a[:,1]))
print(a[(a[:,1]>=50) & (a[:,1]<=90)]) print(a[(a[:,1]>=45) &
```